AI Investment, EU vs US

Who Builds, Funds, and Uses AI — and Where Europe Fits

Project documentation

June 25, 2026

Abstract

This document explains, step by step, the empirical project behind a single claim: *Europe researches and uses artificial intelligence, but does not build, patent, or fund it.* The evidence is a set of directly observable quantities. On the *production* side, the United States and China account for virtually all frontier AI—models, leading developers, patents, compute, and data centres—while Europe is near-absent. On the *research* side, Europe is genuinely strong: it is second in the world in AI research citations. On the *use* side, European firms and citizens adopt AI heavily and fast. The result is a paradox: Europe does the science and the consuming, and even leads on trust and regulation, while the capital and the companies that own AI sit elsewhere. The document covers (i) the reproducible two-stage pipeline, (ii) every data series with its source, units, and treatment, and (iii) the eleven descriptive charts that carry the argument. The project is *descriptive by design*: these are measured magnitudes, not a latent causal channel, so it runs no econometrics. The adoption series is pulled live from Eurostat; the production, investment, infrastructure, public-money, and talent figures are read from the Stanford AI Index 2026 and should be refreshed each report cycle.

Contents

1 Overview and research question

Thesis. A common framing of the EU's position in AI fixates on the venture-capital gap with the United States. This project widens and sharpens the point into a structural contrast across the whole AI value chain. The claim breaks into five legs, each mapped to one or more charts:

(A) **Funding.** Private capital for AI is overwhelmingly American, and the lead is *accelerating*, not closing.

(B) **Production.** The frontier models, the leading developers, the patents, the data centres, and the chip supply chain are American or Chinese; no European company is among the top model builders.

(C) **Research.** Yet Europe is strong *upstream*: it is second in the world in AI research citations and a major publisher. The science is European; the products are not—the project's central paradox.

(D) **Use.** European firms and citizens *adopt* AI heavily and fast. Europe is a large, growing customer for AI it did not build.

(E) **Within-Europe divides & talent.** Adoption is uneven (north-west vs south-east; large firms vs SMEs), and Europe's big economies are weak magnets for AI talent even though several hold talent well.

Why descriptive, not econometric. The sibling precautionary-saving project devotes most of its length to isolating a *latent* causal channel, because there co-movement is not causation. Here the situation is different: every leg above is a *directly measured* quantity—a count of models, a flow of dollars, a share of firms, a count of data centres. The argument is an accounting comparison across the value chain, not an identification problem. There is no hidden variable to recover and nothing to instrument. The honest altitude is therefore a set of clean, well-sourced charts; what the project owes the reader is rigour about *data provenance and comparability*, which Section 3 supplies.

What the documentation covers. Section 2 describes the pipeline. Section 3 justifies every data choice and states the comparability caveats that matter. Section 4 walks through the eleven charts in narrative order. Sections 5–6 synthesise and qualify the findings. The appendix lists the file tree, commands, and a data dictionary.

2 The reproducible pipeline

The project is split into two scripts by *function*, so each stage can be run, audited, and re-run independently. Data collection (network-bound, non-deterministic across vintages) is separated from figure rendering (fast, deterministic, offline). There is deliberately no third stage: per Section 1 the analysis is descriptive.

Stage / script	Reads	Writes
`01_collect_data.py`	Eurostat (live API) + AI Index 2026 figures	`data/*.csv` (15 tidy series)
`02_make_figures.py`	`data/*.csv`	`figures/*.png` (11 charts, A–G)

One-command reproduction. The wrapper `run_all.sh` installs the pinned dependencies and runs the two stages in order:

```
bash run_all.sh        # pip install, then run 01 -> 02 (classic Python)
```

Classic Python, no virtual environment. The project runs directly on a standard system Python ($\geq$ 3.11). `run_all.sh` installs the requirements into the chosen interpreter (`python3` by default; override with `PYTHON`) and then runs the two stages. Dependencies are pinned in `requirements.txt`, so the charts are reproducible across machines.

Network robustness. The adoption data is pulled through the keyless `eurostat` package. Each pull is wrapped in `try/except`: a failed source prints a clear message and leaves any CSV from a previous run untouched. The latest survey year is detected automatically from the returned table, so a new Eurostat release flows through to the country and firm-size charts with no code change.

3 Data

For each series we state the *source* and dataset code, the *units*, the *coverage*, and any *transformation* and *caveat*. The adoption series is free and keyless; the AI Index figures are public but, as the report's underlying data is proprietary (Quid, Epoch AI, etc.), are read from the report PDFs and entered as documented constants with their figure numbers—the project's one set of "verify before publishing" numbers.

3.1 Use: AI adoption by EU enterprises

Source / code
Eurostat `ISOC_EB_AI`, indicator `E_AI_TANY` (enterprises using at least one AI technology), unit `PC_ENT`, size class `GE10` (10+ persons employed). Pulled live.

Coverage / units
Annual; available years 2021, 2023, 2024, 2025 (no 2022). Percentage of enterprises using ≥ 1 AI technology—an *extensive* measure (any use), not intensity or spend.

Caveat: series break
Eurostat broadened the surveyed AI-technology list between the 2023 and 2024 waves; the EU-27 rate jumps 8.1% $\rightarrow$ 13.5%, partly a definitional effect. The 2024 $\rightarrow$ 2025 step (13.5% $\rightarrow$ 19.9%) is on the consistent definition. The trend chart shades the break.

Firm size
The same series for `EU27_2020`, split by size classes `10-49`, `50-249`, `GE250`.

3.2 Funding: private AI investment by region

Source
Stanford AI Index (2024/25/26 editions); underlying data Quid. Current-USD private investment.

Views
(i) cross-country bar, 2025 (Fig. 4.2.8); (ii) US/Europe/China anchor path 2023–25 (Figs. 4.3.10, 4.2.11); (iii) cumulative 2013–25 (Fig. 4.2.12); (iv) generative-AI split (Fig. 4.2.10).

Caveat
"Europe" is the AI Index regional line and *includes the UK*; the report publishes no clean EU-27 aggregate. China's private figure excludes state "guidance funds" (an estimated US$184 bn, 2000–2023).

3.3 Production: notable models and developers

Source
Epoch AI's curated "notable models" dataset, via AI Index 2026. A model is attributed to a country if ≥ 1 author is affiliated there (double-counting possible); "notable" is a manual curation, so read it as a pattern, not a census.

Series
Models by national affiliation, 2025 (Fig. 1.1.1/1.1.2: US 59, China 35, South Korea 8, "Europe" 2); and by organisation, 2025 (Fig. 1.1.6: OpenAI 20, Google 14, Alibaba 11, . . .). DeepMind is counted under Google.

3.4 Research: publications and citations

Source
AI Index 2026, Ch. 1.6 (OpenAlex-based). Regional *shares* of global AI publications (Fig. 1.6.6) and citations (Fig. 1.6.7), 2024.

Use here
Europe's share at each stage of the pipeline (citations 19.5%, publications 11.1%, patent citations 4.2%, patents 3.0%, models $\approx$ 2%) is assembled into the "paradox" chart. Patent shares are from Ch. 1.7 (PATSTAT/EPO; China 74%, US 12%, Europe 3%).

3.5 Infrastructure: data centres and the chip supply chain

Source
Cloudscene via AI Index 2026 (Fig. 1.3.2): number of data centres by country, 2025 (US 5,427, more than $10\times$ any other).

Ownership
The hardware stack is non-European: chips *designed* by Nvidia (US), *fabricated* by TSMC (Taiwan, a near-single point of dependency) and Samsung (Korea), with memory from SK Hynix/Samsung (Korea). Counts do not capture facility size or capacity; the chart's caption says so.

3.6 Public money and talent

Public investment

AI Index 2026, Ch. 8.5. Cumulative 2013–24 government AI spending: US ≈$20.5 bn (grants $15.9 + contracts $3.9 + OTAs $0.7; DoD = 73% of contracts) vs European contracts ≈$3.7 bn (UK $1.6, Germany $0.5, France $0.3). **Caveat:** US and Europe are measured differently (FPDS obligations vs TED contract ceilings) and presented separately; the only like-for-like is contracts (US ≈$4.6 bn vs Europe ≈$3.7 bn).

Talent

LinkedIn via AI Index 2026: net AI talent migration per 10,000 members, 2025 (Fig. 4.4.23), and talent concentration (Fig. 4.4.21). Context: inflow of AI researchers/developers to the US fell 89% since 2017.

3.7 Two comparability caveats that matter

(i) **Incommensurable units, kept separate.** Adoption (a share of firms), investment (a flow of dollars), models (a count), and citations (a share) are never combined into one index; they are parallel lenses on the same gap. In particular, Eurostat firm adoption (≈ 20%) and the McKinsey "organisational adoption" figure (88%) measure different things and must be cited separately.

(ii) **"Europe" is defined differently across sources.** In investment it includes the UK; in models/research it is a regional aggregate; in data centres and talent it appears as individual countries. Each chart's caption states which.

4 Descriptive evidence

The eleven charts are intentionally simple; in narrative order—fund, build, research, house, publicly fund, use, staff—they are the argument.

The funding gap.

The production gap.

The paradox: strong in science, absent in products.

The physical backbone and the public purse.

The consumer side: Europe uses AI heavily.

Talent.

5 Interpretation

Putting the legs together:

- **Funding (A).** Unambiguous and widening. US private AI investment more than doubled in 2025 to US$285.9 bn; cumulatively since 2013 the US has attracted ≈US$757 bn against Europe's low tens of billions, and even public AI money is lopsided.

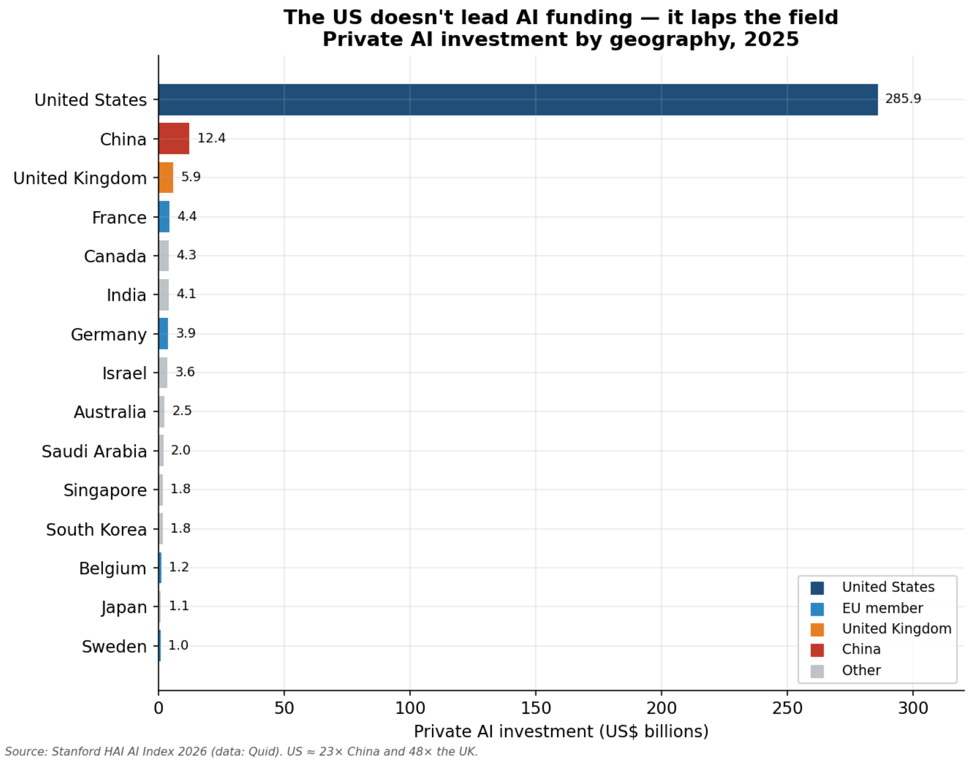

Figure 1: Leg A. Private AI investment by geography, 2025. The US bar runs off the field on purpose: at US$285.9 bn it is ≈ 23× China and 48× the UK. Europe's leaders are slivers.

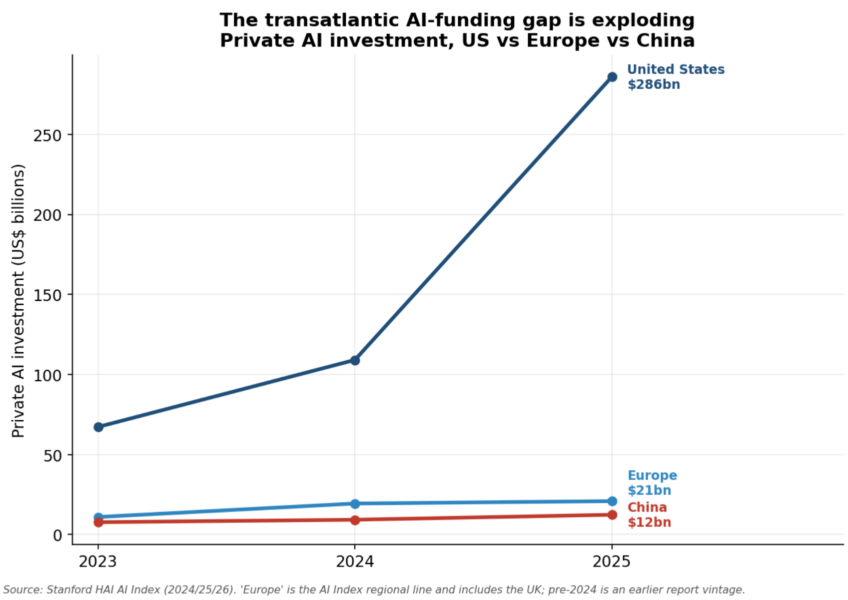

Figure 2: Leg A. US vs Europe vs China, 2023–25. The US curve goes vertical (+≈ 160% since 2024) while Europe (+≈ 7%) and China stay near the floor. "Europe" includes the UK; pre-2024 is an earlier report vintage.

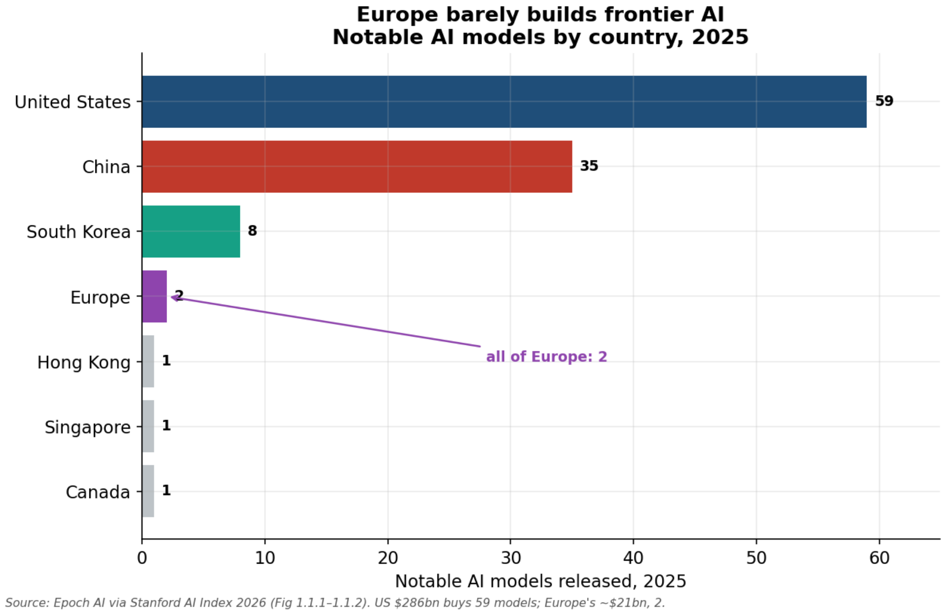

Figure 3: Leg B. Notable AI models by country, 2025 (Epoch AI). The US released 59 and China 35; *all of Europe* released 2. Put beside the funding bar: US $286 bn buys 59 models; Europe's ≈$21 bn, two.

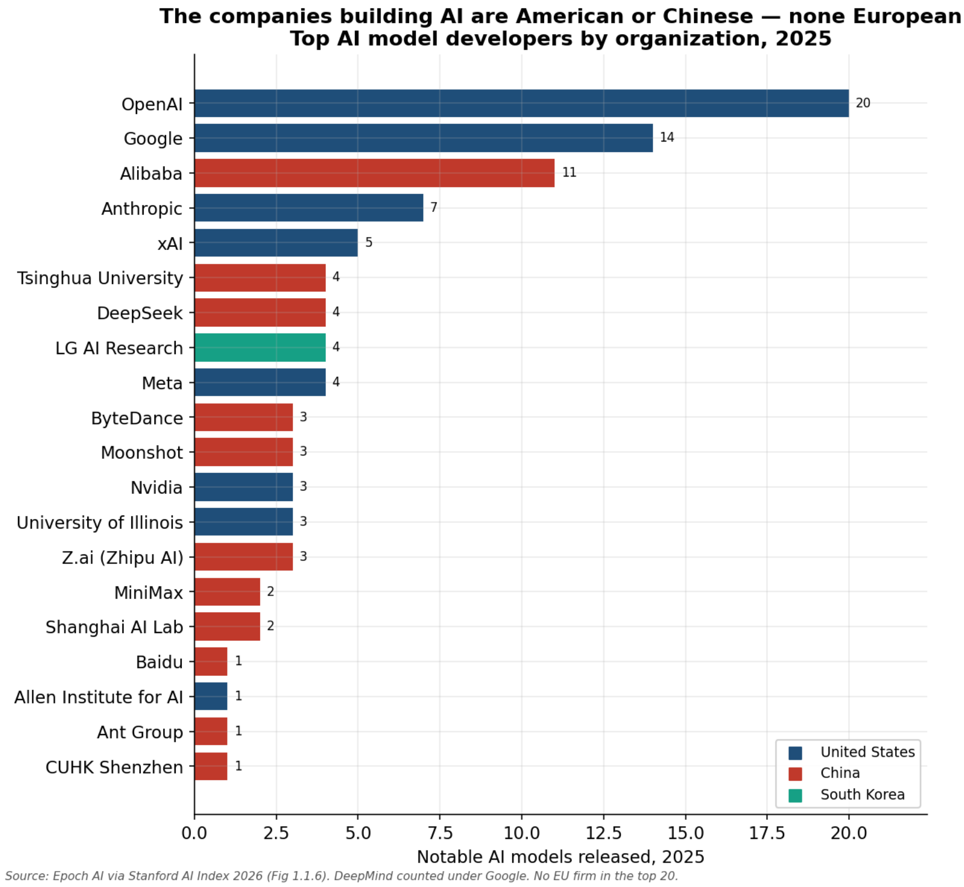

Figure 4: Leg B. The organisations building frontier models, 2025, coloured by origin. Every one of the top twenty is American, Chinese, or (for LG) South Korean. *No European company appears.*

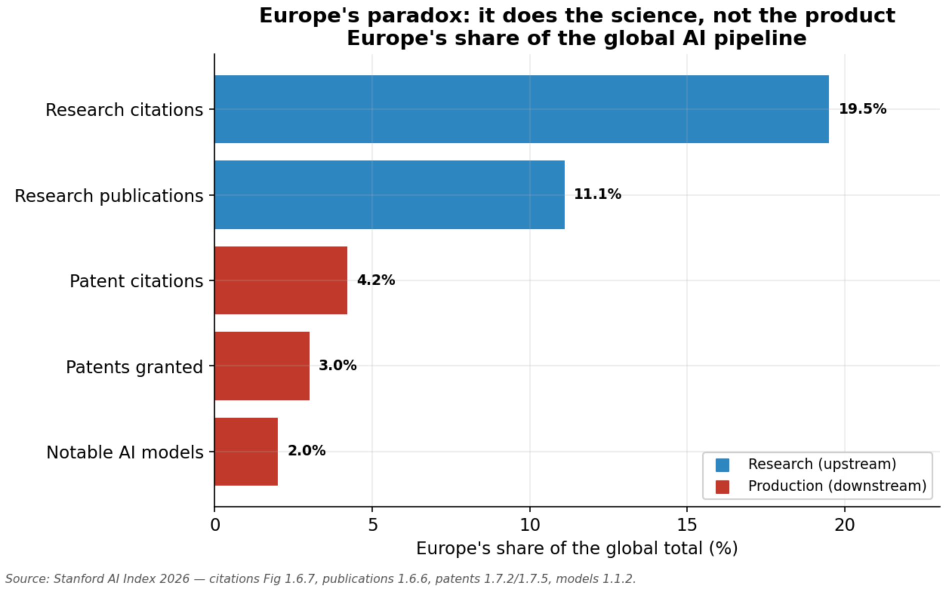

Figure 5: Leg C. Europe's share of the global AI pipeline. Upstream (research) it is large—19.5% of citations, 11.1% of publications; downstream (production) it collapses—3% of patents, $\approx$ 2% of notable models. Europe does the science, not the product.

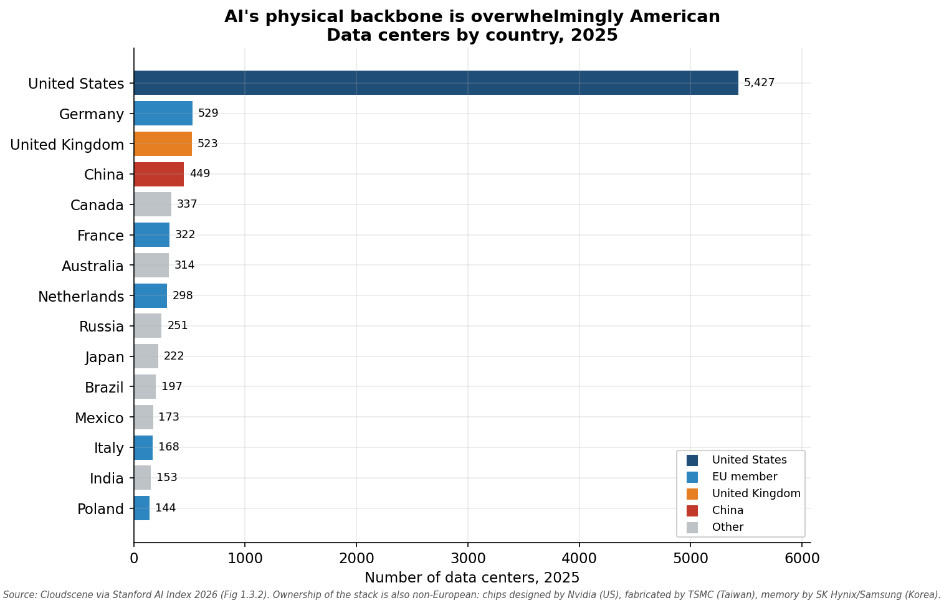

Figure 6: Leg B. Data centres by country, 2025 (Cloudscene). The US hosts 5,427, more than 10× any other country; ownership of the chip stack (Nvidia/TSMC/SK Hynix) is likewise non-European. Counts ignore facility size.

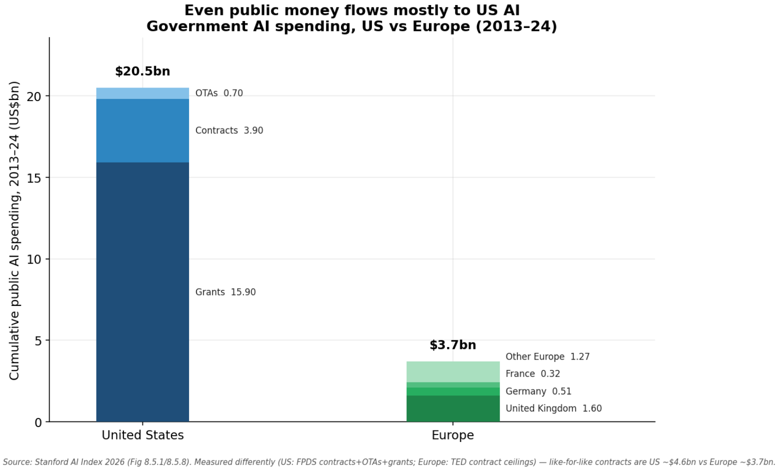

Figure 7: Leg A. Cumulative government AI spending, 2013–24. Even public money is lopsided: US ≈$20.5 bn vs European contracts ≈$3.7 bn. The two are measured differently (caption); like-for-like contracts are US ≈$4.6 bn vs Europe ≈$3.7 bn.

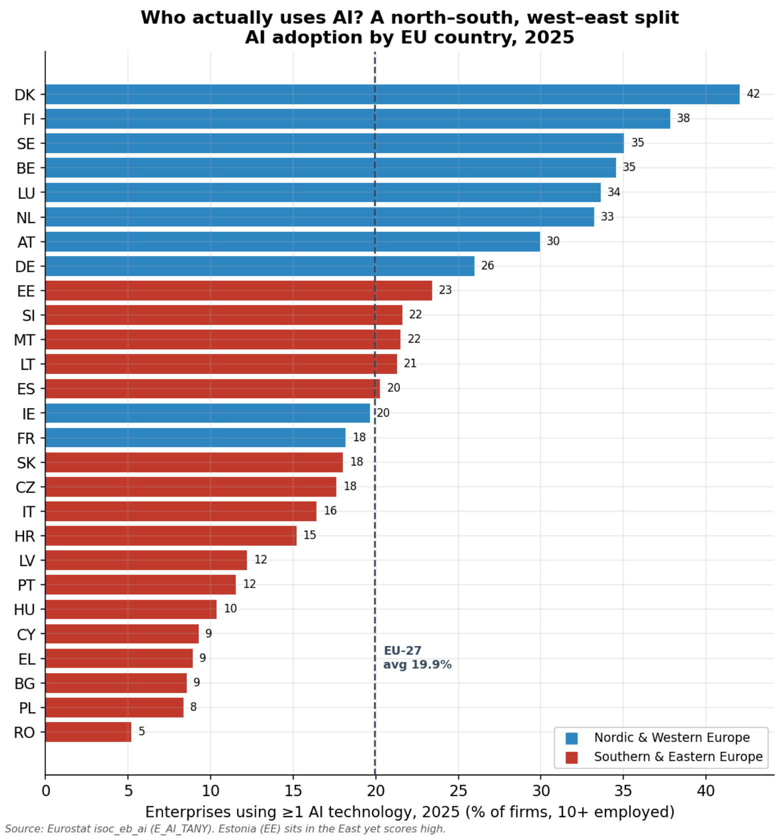

Figure 8: Leg D/E. AI adoption by EU country, 2025 (Eurostat). A north-west/south-east split: Denmark 42% leads, Romania 5% trails—an eight-fold range around the 19.9% EU-27 average.

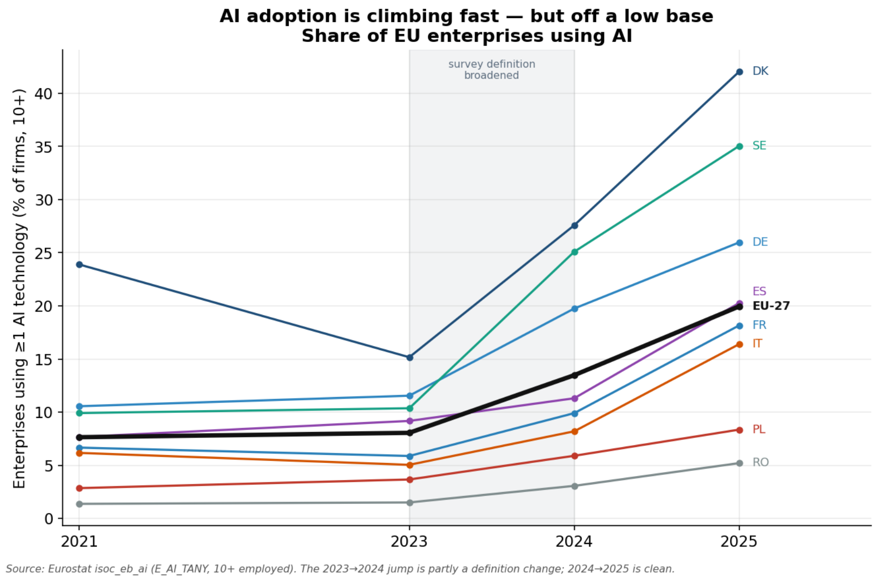

Figure 9: Leg D. AI adoption over time. The EU-27 rate roughly doubled in two years. The 2023 $\rightarrow$ 2024 segment is shaded because the survey definition was broadened; 2024 $\rightarrow$ 2025 is clean.

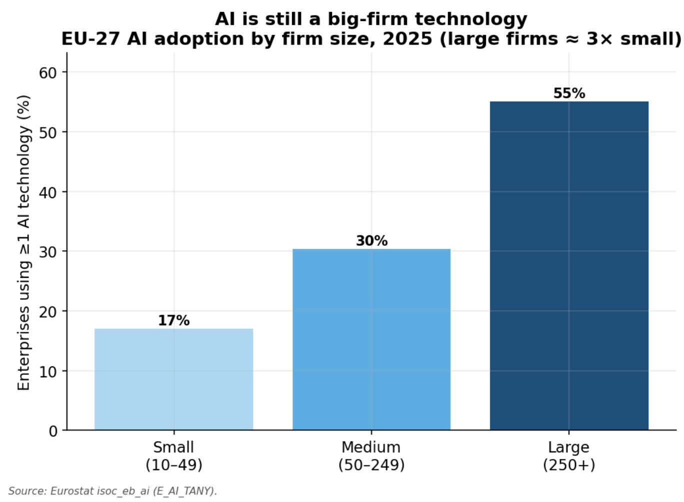

Figure 10: Leg E. EU-27 AI adoption by firm size, 2025. Large firms (55%) adopt at roughly three times the small-firm rate (17%): the SME gap within Europe.

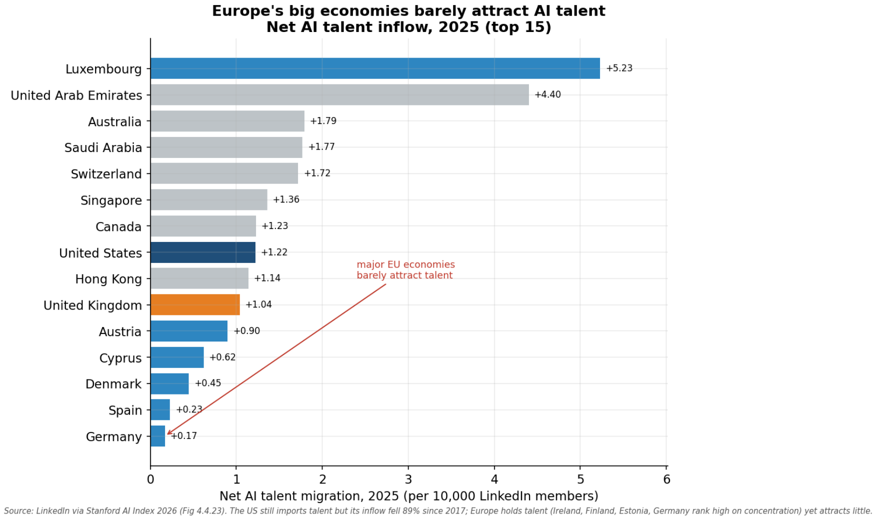

Figure 11: Leg E. Net AI talent migration, 2025 (LinkedIn). Small hubs (Luxembourg) attract strongly, but Europe's big economies (Germany +0.17, Spain +0.23) barely do. The US still imports talent (+1.2) though its inflow fell 89% since 2017.

- **Production (B).** Starker still. Europe built two notable models in 2025, holds 3% of AI patents, and places no company in the top twenty developers; the data centres and the chip supply chain are American, Chinese, Taiwanese, and Korean.
- **Research (C).** The twist: Europe is genuinely strong upstream (19.5% of AI citations, second only to China), and its papers come disproportionately from academia. The science is European; the commercialisation is not.
- **Use (D) and divides (E).** Europe adopts AI heavily and fast—the EU-27 firm rate doubled to $\approx 20\%$ in two years, and population-level use is high—but unevenly (Denmark 42% vs Romania 5%; large firms $\approx 3\times$ SMEs), and its big economies attract little AI talent.

The synthesis is the headline: *Europe is integrating AI into its economy, and doing much of the underlying science, while the capital, the companies, and the infrastructure that* own *AI accrue almost entirely elsewhere.* Europe's distinctive strengths—research and, separately, trust and regulation—sit upstream and sideways of the value chain, not at its commercial core. The adoption gap is closing; the production and funding gaps are not.

6 Limitations and caveats

- **Adoption series break.** The 2023 $\rightarrow$ 2024 jump partly reflects a broadened survey definition; lead with 2024 $\rightarrow$ 2025 or annotate the break.
- **"Europe" is not one thing.** It includes the UK in investment, is a regional aggregate in models/research, and appears as individual countries in data centres and talent (Sec. 3.7).
- **"Notable models" is a curation.** Epoch AI's set is manual and author-affiliation-based (double-counting possible); read it as a pattern.
- **Data-centre counts $\neq$ capacity.** Cloudscene counts facilities, not size, power, or utilisation.
- **Public-spend methodologies differ.** US (FPDS obligations) and Europe (TED contract ceilings) are not directly comparable beyond contracts.

- **Talent nuance.** The 2025 data show Europe's big economies as weak *magnets*, not a dramatic outflow; the sharp move is the 89% fall in US inflow. Do not overclaim a European "brain drain."
- **Two adoption scales.** Eurostat firm adoption ($\approx$ 20%) and McKinsey organisational adoption (88%) measure different things.
- **Vintages.** The adoption series is pulled live and revised by Eurostat; the AI Index figures are hand-entered from a periodically updated report. Regenerate and re-verify before citing any number.

A Reproduction details

A.1 File tree

```
01_collect_data.py     # STEP 1  data collection -> data/*.csv (15 series)
02_make_figures.py     # STEP 2  figures         -> figures/*.png (11 charts)
run_all.sh             # pip install + run 01 -> 02 (classic Python, no venv)
requirements.txt       # pinned dependencies
data/                  # generated tidy CSVs
figures/               # generated charts
README.md              # short project readme
documentation.tex      # this document
```

A.2 Commands

```
# one-shot, fully reproducible
bash run_all.sh

# or stage by stage (after: pip install -r requirements.txt)
python 01_collect_data.py
python 02_make_figures.py
```

A.3 Data dictionary (generated CSVs)

file	shape	contents
`ai_adoption_by_country.csv`	latest yr	geo, value (%), year
`ai_adoption_trend.csv`	long panel	geo, year, value (%)
`ai_adoption_by_firmsize.csv`	latest yr	size_emp, value (%), year
`ai_investment_by_geo.csv`	2025	country, investment ($bn), year
`ai_investment_us_eu_cn.csv`	2023–25	year, US, Europe, China, source
`ai_investment_cumulative.csv`	2013–25	country, cumulative ($bn)
`ai_investment_genai.csv`	2025	region, gen-AI ($bn)
`ai_models_by_country.csv`	2025	country, models, year
`ai_models_by_org.csv`	2025	organization, models, country
`europe_share_pipeline.csv`	latest	stage, Europe share (%), kind, source
`research_by_region.csv`	2024	region, publications %, citations %
`datacenters_by_country.csv`	2025	country, data centres, year
`public_ai_investment.csv`	2013–24	region, category, $bn
`talent_concentration.csv`	2025	country, concentration (%)
`talent_net_migration.csv`	2025	country, net per 10k

The charts in this document come from a clean end-to-end run and are reproducible via `run_all.sh`. The adoption series is pulled live from Eurostat; the production, investment, infrastructure, public-money, and talent figures are read from the Stanford AI Index 2026. Regenerate the data and re-compile before citing.